INCH-POUND
MIL-DTL-24643/48F
1 October 2009
SUPERSEDING
MIL-DTL-24643/48E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 1000 VOLTS, TYPE LSDNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for thickness). Standard identification code shall be applied by Method 1 or Method 3. If identification Method 1 is used, the insulation color shall be black.
Fourth	-	Two conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/48	Type and size	AWG conductor size	Insulation thickness min. avg. (inch)	Cable jacket thickness (min. avg.) (inch)	Overall diameter	
					min. (inch)	max. (inch)
-01UN	LSDNW-3	16 (Class B)	0.030	0.040	0.361	0.390
-02UN	LSDNW-4	14 (Class B)	0.030	0.040	0.398	0.430
-03UN	LSDNW-9	10 (Class B)	0.030	0.040	0.504	0.545
-04UN	LSDNW-14	9 (Class B)	0.045	0.040	0.564	0.610
-05UN	LSDNW-23	7 (Class B)	0.045	0.045	0.638	0.690
-06UN	LSDNW-50	3 (Class C)	0.045	0.050	0.842	0.910
-07UN	LSDNW-75	1 (Class C)	0.055	0.055	1.000	1.080
-08UN	LSDNW-100	0 (Class D)	0.055	0.055	1.082	1.170
-09UN	LSDNW-200	0000(Class D)	0.055	0.070	1.498	1.413
-10UN	LSDNW-300	300MCM (Class D)	0.055	0.085	1.756	1.671
-11UN	LSDNW-400	400 (127) (Navy std)	0.055	0.085	1.984	1.899

TABLE II. Electrical details.

Type and size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand min. (volts)	Accelerated service loading current (amperes)
LSDNW-3	4.3	6000	4000	NA
LSDNW-4	2.68	5000	4000	NA
LSDNW-9	1.08	3800	4000	NA
LSDNW-14	0.86	4400	5500	100
LSDNW-23	0.54	3700	5500	130
LSDNW-50	0.21	2600	5500	215
LSDNW-75	0.13	2600	7000	285
LSDNW-100	0.11	2400	7000	335
LSDNW-200	0.053	2400	7000	525
LSDNW-300	0.0377	2400	7000	700
LSDNW-400	0.0273	2400	7000	835

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REQUIREMENTS:	
Qualification required.	
INSPECTION:	
Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table II
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	See table II
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	(see table II)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (pounds per inch thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Permanence of printing (insulation), Method 1 only (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Accelerated service, Sizes 14 and larger (see table II for load current)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSDNW	3	2500
LSDNW	4 through 23	2000
LSDNW	50	1500
LSDNW	75 and 100	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-052)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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